

1. Machine Learning

Machine Learning is a part of Artificial Intelligence where computers learn from data and improve their performance without being explicitly programmed. It helps systems to find patterns and make predictions automatically. It is widely used in real life applications like recommendation systems, fraud detection, healthcare, and search engines.

2. Linear Regression

Linear Regression is a supervised learning algorithm used to predict continuous numerical values. It finds the relationship between input variables and output by fitting a straight line. It is commonly used for predicting marks, prices, and trends based on past data.

3. Features

Features are the input variables used by a machine learning model to make predictions. They provide information to the model. In this dataset, Study_Hours, Attendance, and Assignments are features.

4. Labels

Labels are the output values that the model tries to predict. They are the final result. In this dataset, Marks is the label because it is what we want to predict.

5. Training vs Testing Data

Training data is used to teach the model patterns from the dataset. Testing data is used to check how well the model performs on unseen data. This helps to evaluate the accuracy of the model.

6. Real Life Applications

Machine Learning is used in many real-world applications such as YouTube recommendations, Google Maps navigation, online shopping suggestions, fraud detection in banks, and medical diagnosis systems in healthcare.